

Project-Agricultural Chickens Production Pennsylvania Affected By COVID-19

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```
Registered S3 method overwritten by 'quantmod':
```

```
  method      from  
as.zoo.data.frame zoo
```

```
#Pre-Processing
```

Observed that variables Value and Period are both character columns, not numeric. Conversion to numeric process and remove commas present.

```
[1] "numeric"
```

```
[1] 40009000 51468000 48643000 42170000 49237000 45623000
```

Confirming no missing values present.

```
[1] FALSE
```

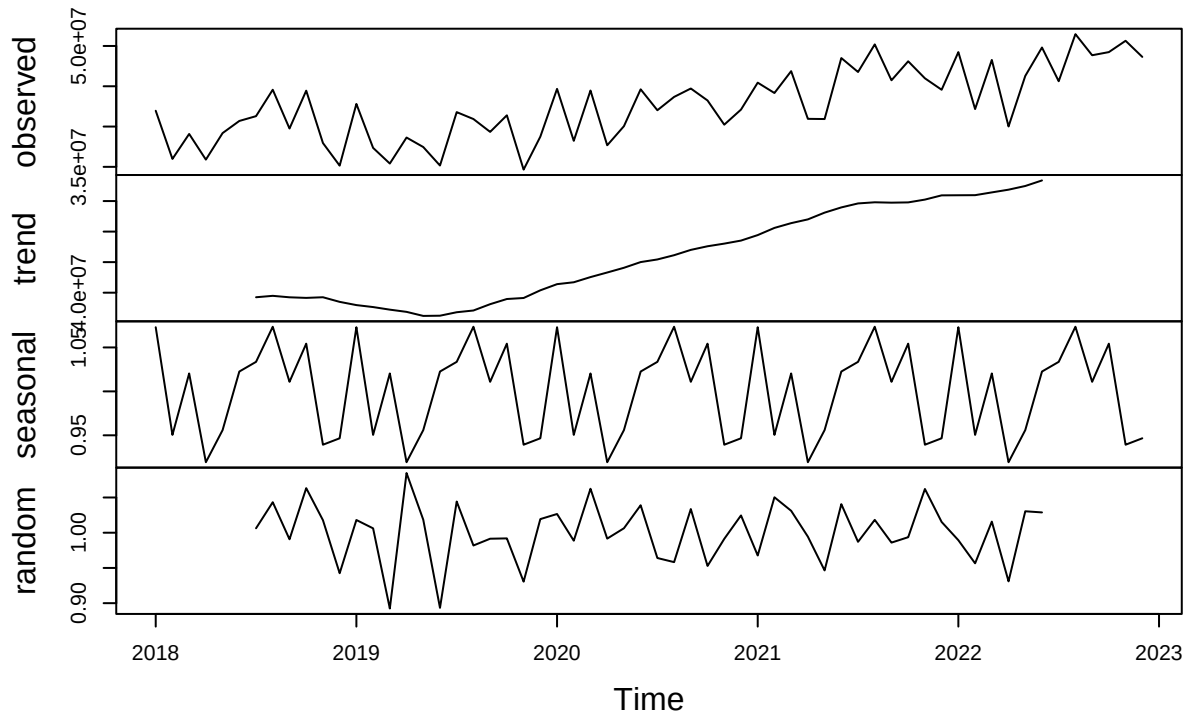
Variable Period converts months to numeric by three-letter format, and the month-letters are all capital. It says for each element in Period, match its position in table in toupper(month.abb), which in turn is saying convert the list month.abb(c(Jan, Feb,...,Dec)) to uppercase, that way it matches the capitalized months in Period.

Sorting, which is important for time order.

```
Time-Series [1:60] from 2018 to 2023: 41962000 35982000 39083000 35896000 39190000 ...
```

```
#Seasonal Decomposition Multiplicative Model
```

Decomposition of multiplicative time series



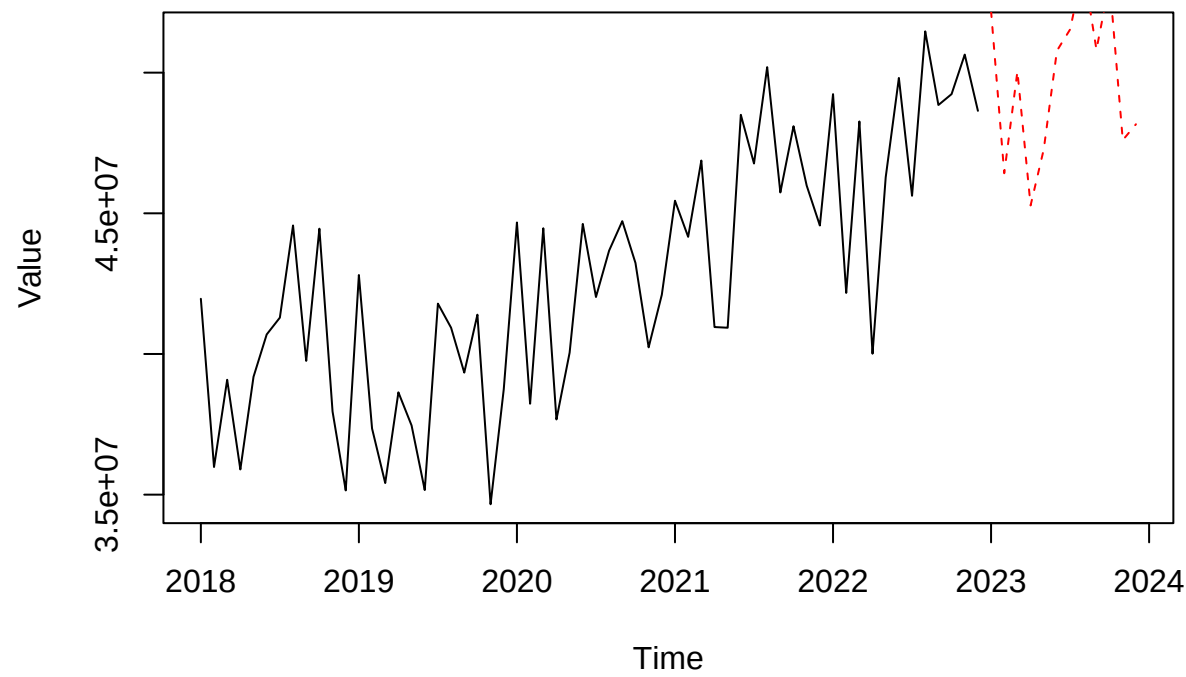
```
[1] 1.0730029 0.9504886 1.0203860 0.9193926 0.9559208 1.0225593 1.0335956
[8] 1.0735540 1.0108901 1.0542736 0.9393133 0.9466229
```

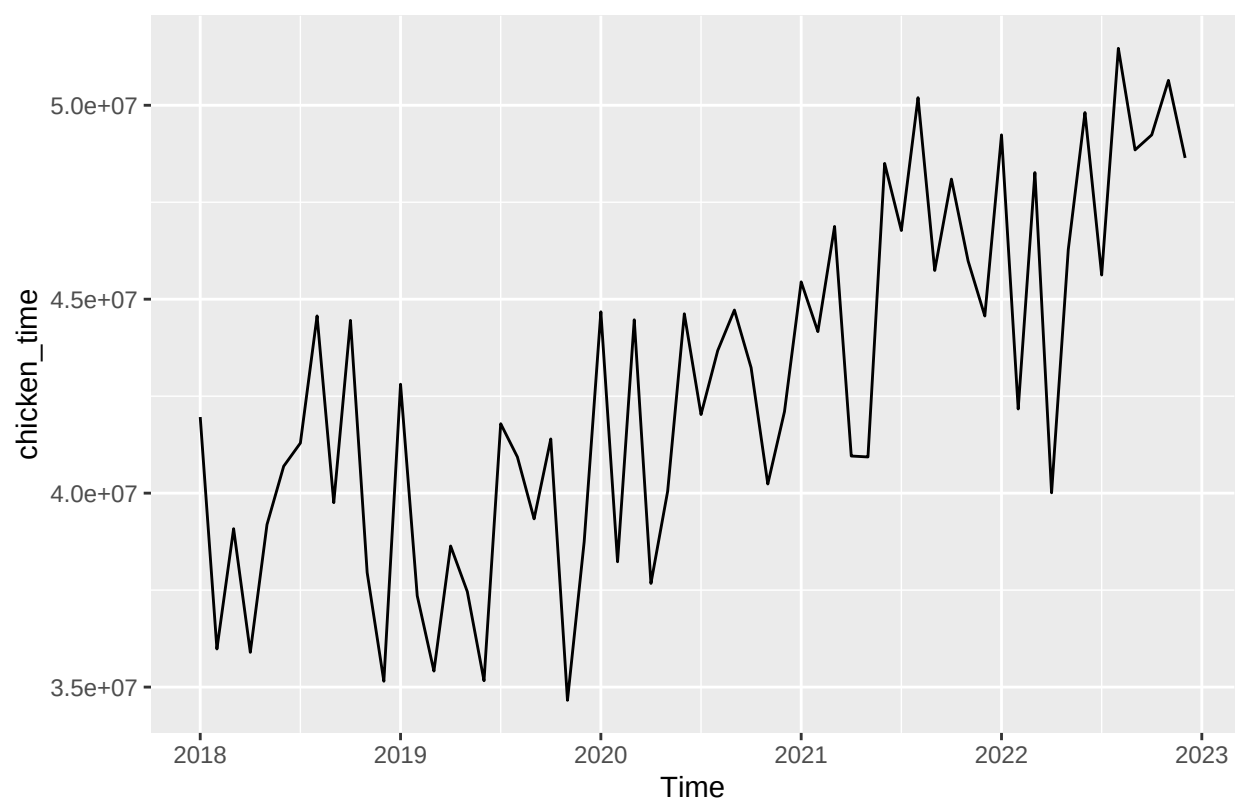
	Jan	Feb	Mar	Apr	May	Jun	Jul
2018	1.0730029	0.9504886	1.0203860	0.9193926	0.9559208	1.0225593	1.0335956
2019	1.0730029	0.9504886	1.0203860	0.9193926	0.9559208	1.0225593	1.0335956
2020	1.0730029	0.9504886	1.0203860	0.9193926	0.9559208	1.0225593	1.0335956
2021	1.0730029	0.9504886	1.0203860	0.9193926	0.9559208	1.0225593	1.0335956
2022	1.0730029	0.9504886	1.0203860	0.9193926	0.9559208	1.0225593	1.0335956

	Aug	Sep	Oct	Nov	Dec
2018	1.0735540	1.0108901	1.0542736	0.9393133	0.9466229
2019	1.0735540	1.0108901	1.0542736	0.9393133	0.9466229
2020	1.0735540	1.0108901	1.0542736	0.9393133	0.9466229
2021	1.0735540	1.0108901	1.0542736	0.9393133	0.9466229
2022	1.0735540	1.0108901	1.0542736	0.9393133	0.9466229

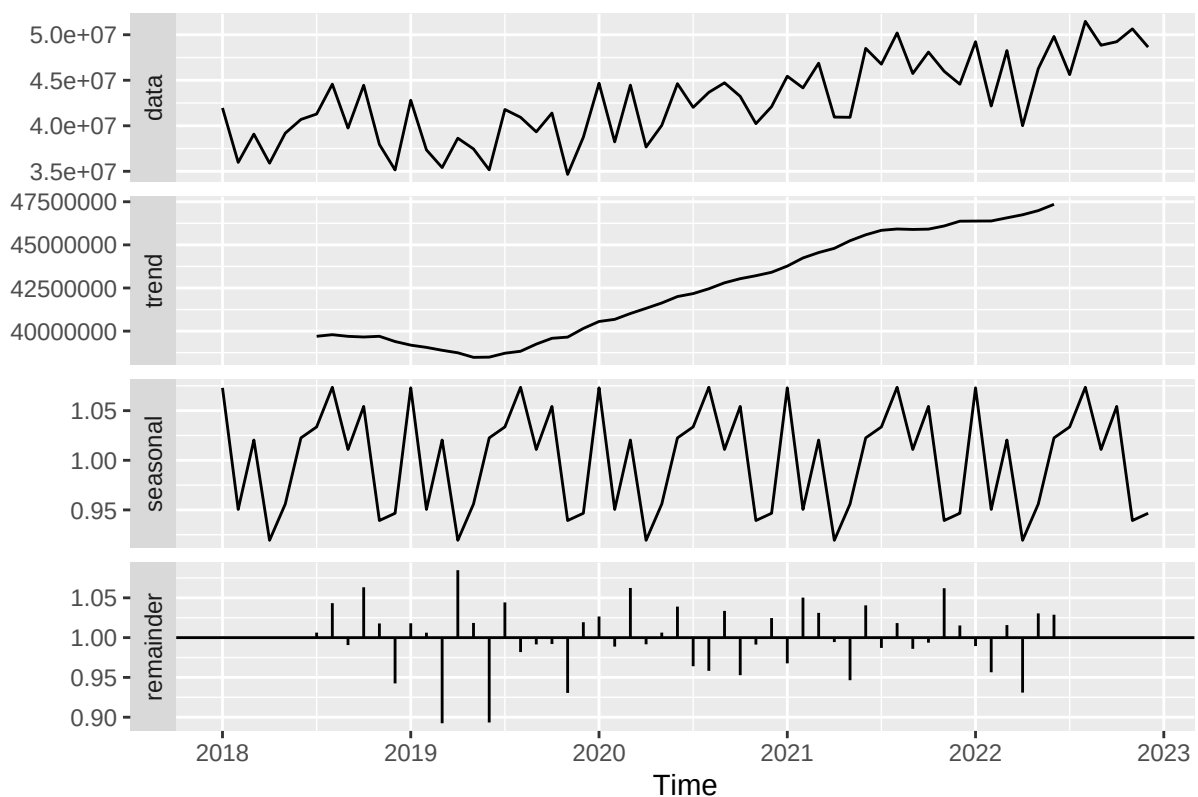
	1	2	3	4	5	6	7	8
52188389	46423489	50045569	45279842	47273866	50778004	51536911	53748325	
	9	10	11	12				
50817243	53213206	47602356	48165912					

Chicken Production Forecast





Decomposition of multiplicative time series



#Holt-Winters Multiplicative Model

Holt-Winters exponential smoothing with trend and multiplicative seasonal component.

Call:

```
HoltWinters(x = chicken_time, seasonal = "multiplicative")
```

Smoothing parameters:

alpha: 0.2110384

beta : 0.0593853

gamma: 0.7322568

Coefficients:

[,1]

a 4.839613e+07

b 2.372964e+05

s1 1.077492e+00

s2 9.554042e-01

s3 1.066137e+00

s4 9.066157e-01

s5 9.985615e-01

s6 1.081984e+00

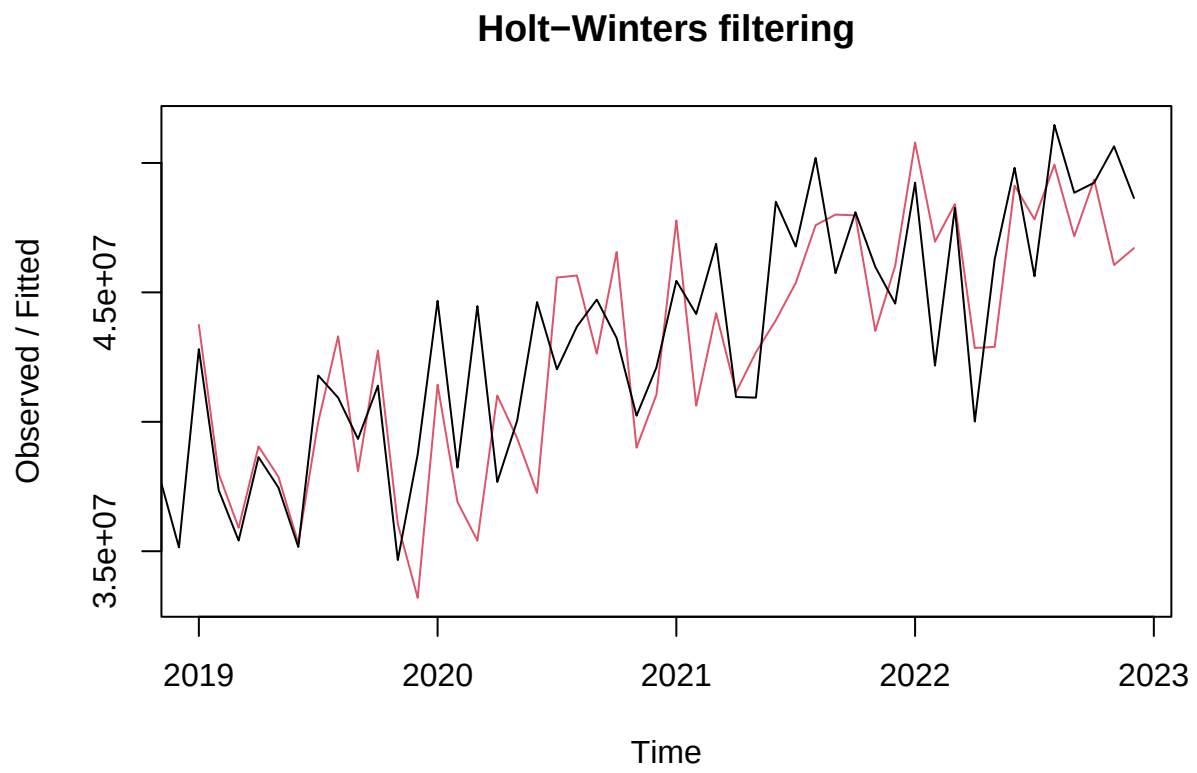
s7 1.010682e+00

s8 1.111475e+00

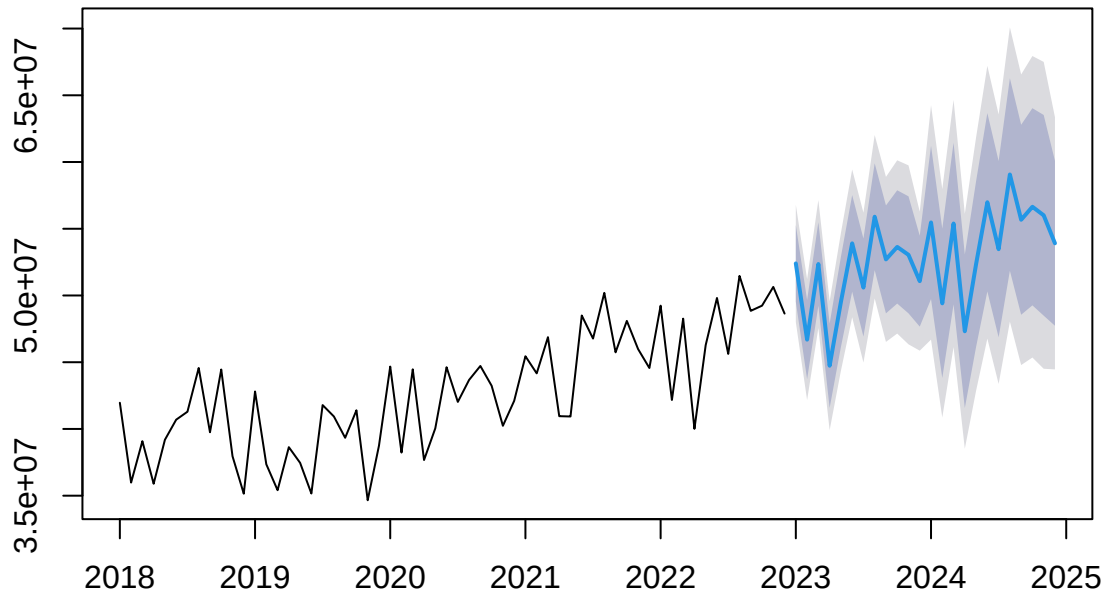
s9 1.042972e+00

s10 1.056502e+00

s11 1.039961e+00
s12 9.966623e-01



Forecasts from HoltWinters



#ARIMA Mixed Model

Series: chicken_time
ARIMA(0,1,2)(1,0,0)[12]

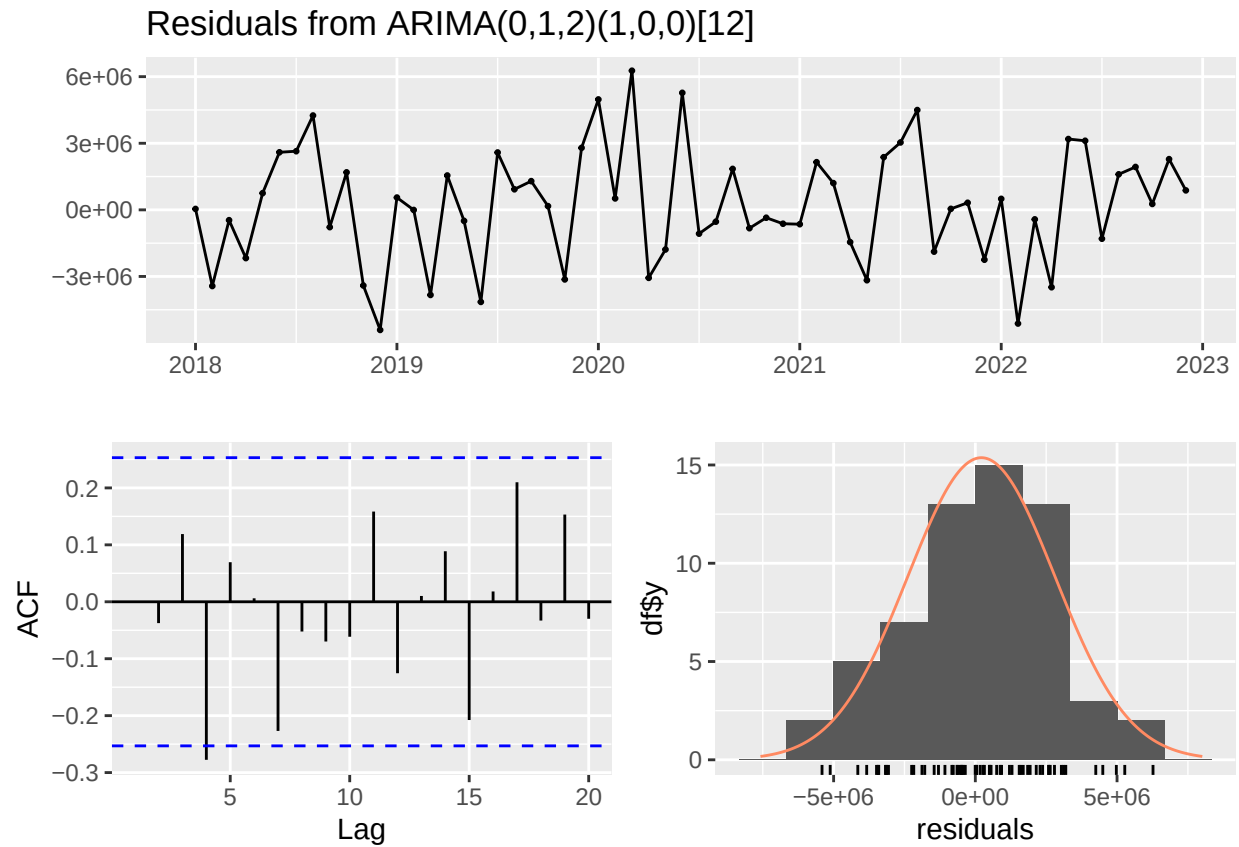
Coefficients:

	ma1	ma2	sar1
	-0.9275	0.3217	0.5945
s.e.	0.1220	0.1665	0.1140

sigma² = 7.161e+12: log likelihood = -958.43
AIC=1924.86 AICc=1925.6 BIC=1933.17

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	213901	2585176	2056380	0.1458906	4.919505	0.6712929	-0.001509238



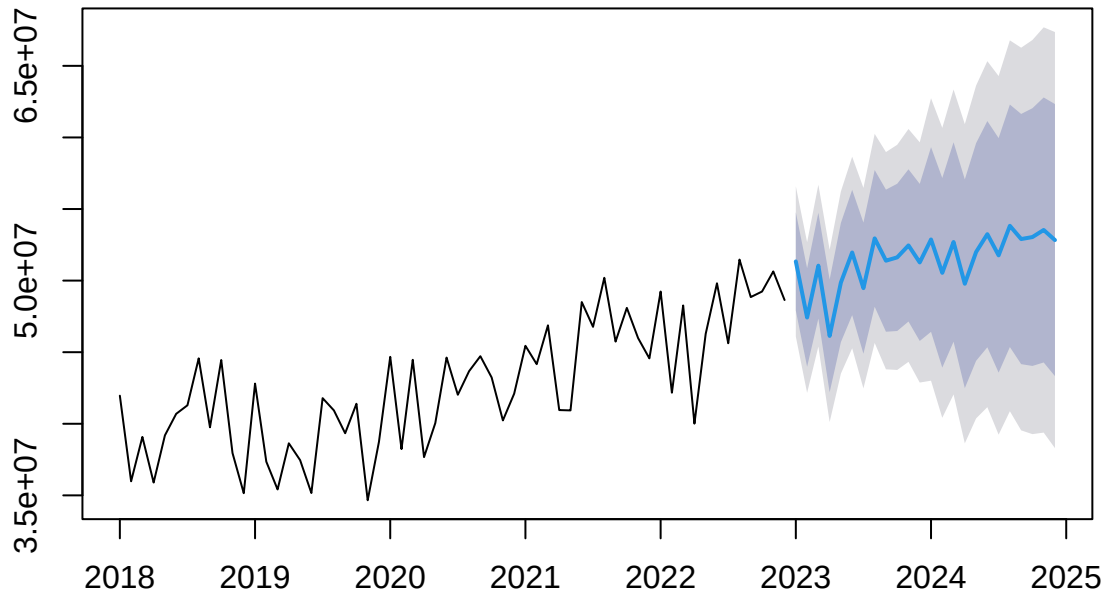
Ljung-Box test

data: Residuals from ARIMA(0,1,2)(1,0,0)[12]

Q* = 14.027, df = 9, p-value = 0.1214

Model df: 3. Total lags used: 12

Forecasts from ARIMA(0,1,2)(1,0,0)[12]



	Point	Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2023		51340117	47910799	54769435	46095428	56584807
Feb 2023		47421397	43983084	50859710	42162951	52679843
Mar 2023		51044573	47350122	54739025	45394397	56694750
Apr 2023		46136789	42202840	50070738	40120334	52153245
May 2023		49865778	45706099	54025457	43504097	56227458
Jun 2023		51964189	47590413	56337964	45275076	58653301
Jul 2023		49474035	44896165	54051905	42472787	56475282
Aug 2023		52948599	48175354	57721844	45648550	60248647
Sep 2023		51391733	46430800	56352665	43804641	58978824
Oct 2023		51622379	46480607	56764152	43758716	59486042
Nov 2023		52457583	47141117	57774048	44326750	60588415
Dec 2023		51269276	45783678	56754874	42879777	59658774
Jan 2024		52872579	46423907	59321250	43010186	62734971
Feb 2024		50543093	43922400	57163786	40417617	60668569
Mar 2024		52696893	45734260	59659525	42048465	63345320
Apr 2024		49779457	42490909	57068005	38632585	60926330
May 2024		51996157	44395656	59596658	40372194	63620120
Jun 2024		53243558	45343413	61143703	41161329	65325787
Jul 2024		51763285	43574453	59952117	39239547	64287023
Aug 2024		53828742	45361059	62296424	40878538	66778945
Sep 2024		52903262	44165623	61640900	39540196	66266327
Oct 2024		53040370	44040869	62039870	39276821	66803918
Nov 2024		53536857	44282901	62790812	39384153	67689560
Dec 2024		52830467	43328869	62332065	38299027	67361907

Forecasts from ARIMA(0,1,2)(1,0,0)[12]

